BeliefRevision · C2

Admit support carefully.
Own the belief decision.

ELI16 · THE IDEA IN PLAIN LANGUAGE

An evidence ledger: a copied report is still the same source, and a calculation does not erase contrary observations.

Why it matters: Numerical revision is meaningful only when the system tracks which evidence contributed, which assumptions were made and which alternatives remain.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

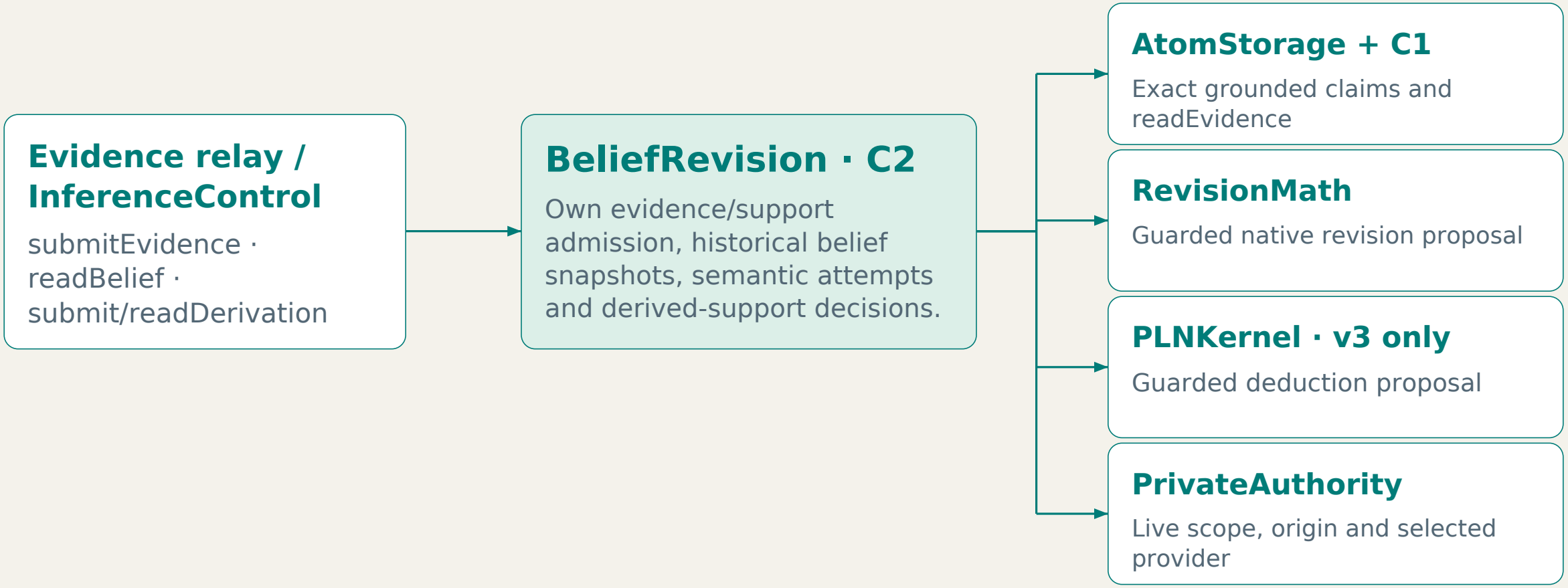
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	**Beliefs · C2**	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Own evidence/support admission, historical belief snapshots, semantic attempts and derived-support decisions.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Support + observation ledger

Accepted occurrence lineage, alternatives and historical snapshots

Semantic attempts

Exact input bindings and durable admission decisions

OUTSIDE THIS RESPONSIBILITY

Original event history

C1 owns episodes and evidence-delivery intent

Pure math and task scheduling

Providers calculate; Attention ranks; C4 budgets work

M2 pln-revision-private/1; fresh M3 pln-derived-private/1 · local v3

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
submitEvidence	Validate C1 evidence or existing support	Admit/duplicate separately from numerical outcome
readBelief	Read exact historical or current snapshot	Aggregate states its admitted coverage
submitDerivation	Bind exact historical premise vector	v3 tagged support; no new observation
readDerivation	Recover a saved semantic decision	Replay does not force another calculation

Bounded local support history; aggregate covers admitted supports, not every retained observation.

A worked example through the boundary

M2 REAL native PLN + DeTTa · generated observations and explicit independence assumptions

01 · STEP

Retain reports

Store original and contrary observations.

02 · STEP

Check lineage

Copied/ancestor support cannot create independent roots.

03 · STEP

Run guarded revision

Eligible exact pair uses the pinned native formula.

04 · STEP

Commit decision

Keep alternatives and explicit aggregate coverage.

What the example establishes

The owner decides what counts; the calculator only proposes.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Admission ≠ aggregation

A report may be processed without changing a numerical truth.

I2

No double evidence

New key or speaker cannot manufacture an independent origin.

I3

Derived support inherits

Deduction carries original roots and ancestry; no fresh observation count.

I4

Owner commits last

Recheck exact bindings, current authority and expected revision before committing.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Proposal computed; owner commit lost	Recover/retry exact attempt under its budget semantics.	Physical calculation may repeat where allowed
Decision saved; relay reply lost	Return the original decision through declared ports.	Duplicate-safe admission
Stale revision / malformed provider response	Refuse or retain explicit outcome; never silently rebind.	CAS and hostile-response tests

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

0.5 / ⅔

Native revision strength / confidence

Recorded M2 generated example; model-specific

2

Retained alternatives in M2 story

Disagreement remains visible

0 new

Observations from derived support

M3 admission invariant, not an accuracy score

These are numerical goldens and lineage/admission results. They do not measure calibrated confidence, predictive accuracy, source independence or per-admission latency.

PROPOSED BENCHMARK · NOT RUN

Proposed: fix support lineage and premise snapshots; vary duplicate ratio, alternatives and CAS contention. Measure admission p50/p95 and verify unchanged evidence-root accounting.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Private SQLite owners, actual guarded native revision/deduction and real DeTTa bytes.
FIXTURE / LOCAL TEST POLICY	Generated claims, declared independence assumptions, malformed proposals and injected crashes.
BLOCKED / UNSUPPORTED	No required supported gate blocked. Existing v2/v3 journals are not migrated or silently widened.
UNRUN	Retraction, arbitrary rule libraries, calibration, hidden-copy detection and distributed belief writers.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure admission costs	Controlled lineage/duplicate/CAS workloads	Latency reported with duplicate and conflict rates
Specify retraction semantics	Versioned provenance and historical-read contract	No erased history or unaccounted evidence
Evaluate numerical usefulness	External task dataset and calibration protocol	Accuracy/calibration separated from conformance

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

The owner decides what counts; the calculator only proposes.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3.py <env> --output <fresh>
```

Uses explicit private profiles and pinned native providers. Historical M2 demo is a CLI observation; separate native Hive cases establish the real-caller gate.

`fusion/beliefs.py`
`fusion/derived_beliefs.py`
`fusion/cognition_contracts.py`
`tests/test_m2.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.